Artificial Neural Networks
Assignment02

Spring 2026

SECTION BAI-6A

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<u>SUBMITTED TO</u>

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<u>SUBMISSION DATE</u>

April 19, 2026

DEPARTMENT OF AI,

FAST -NUCES, KARACHI

Step By Step Explanation and Screen Shots of Outputs:

Figure3 Toy1 Manual:

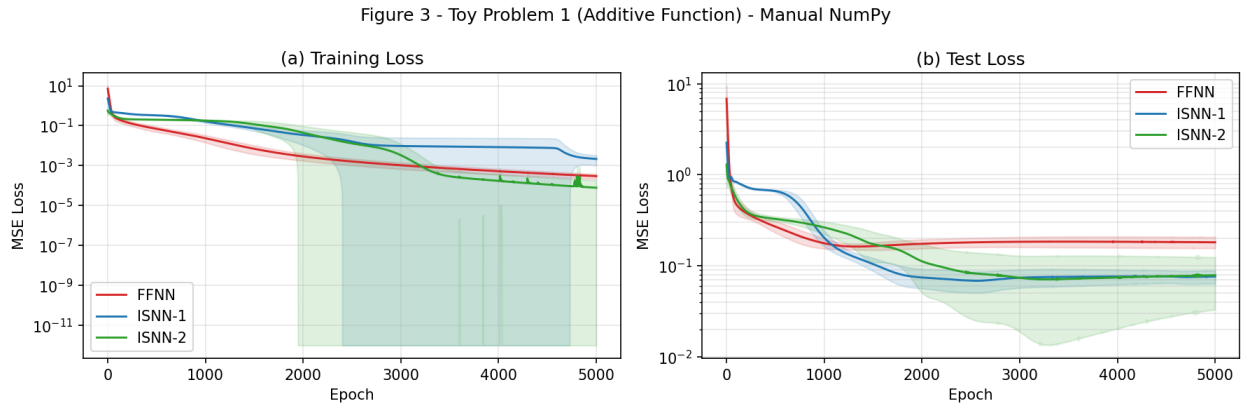


Figure4 Toy1 Manual:

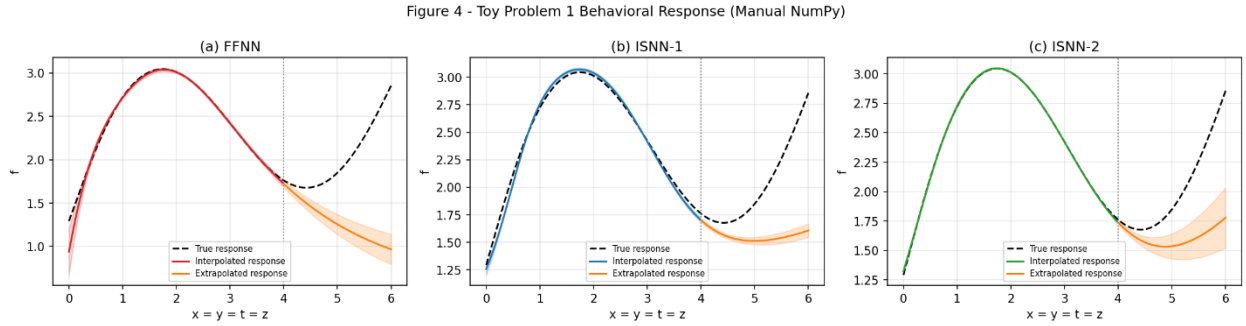


Figure5 Toy1 Manual:

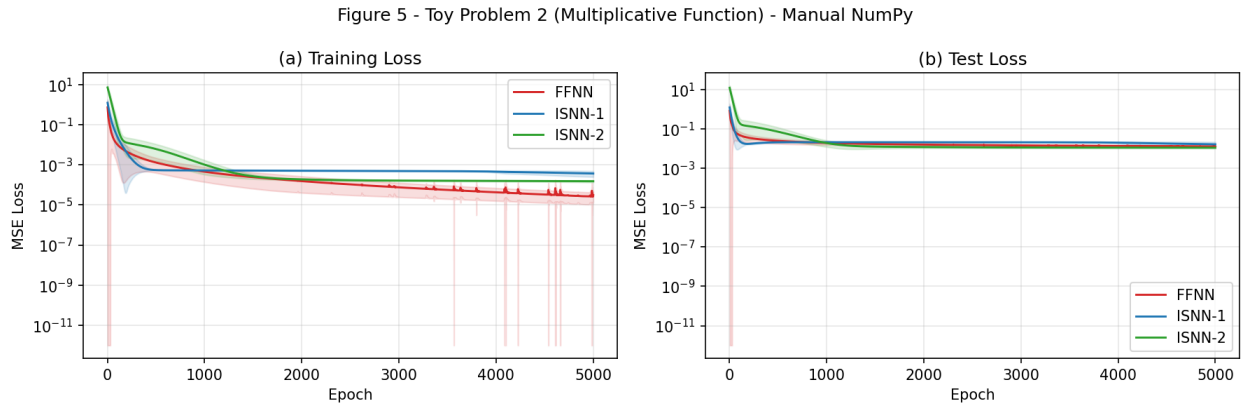


Figure6 Toy1 Manual:

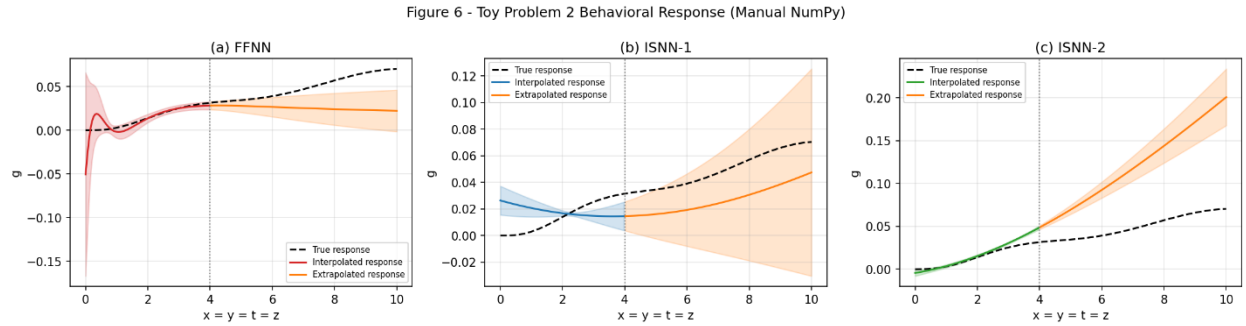


Figure3 Toy1 PyTorch:

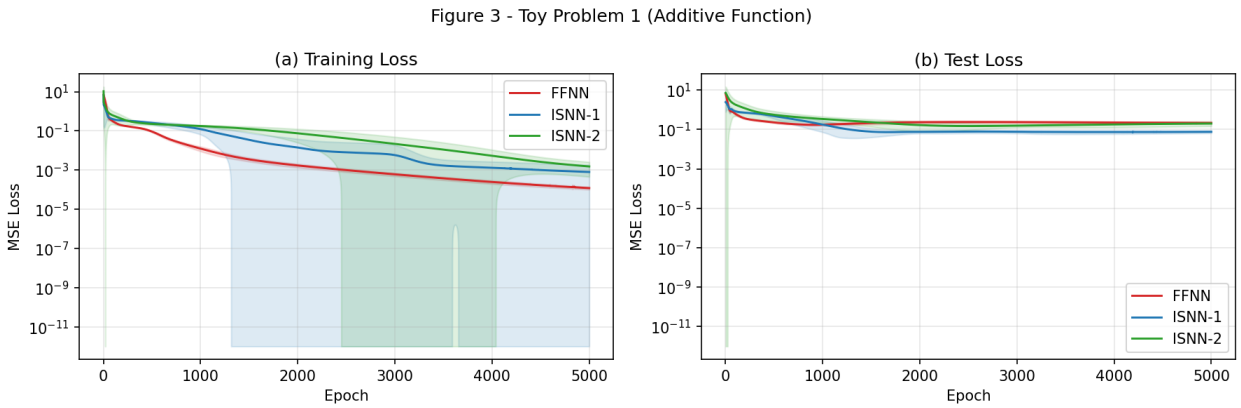


Figure4 Toy1 PyTorch:

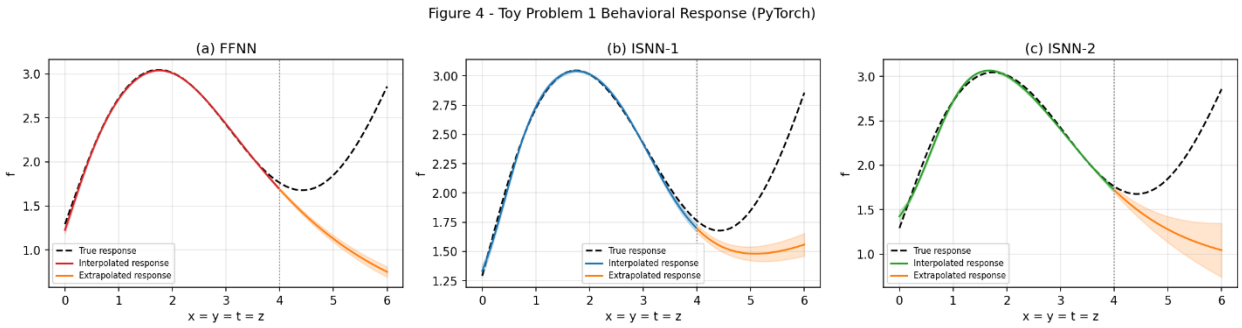


Figure5 Toy1 PyTorch:

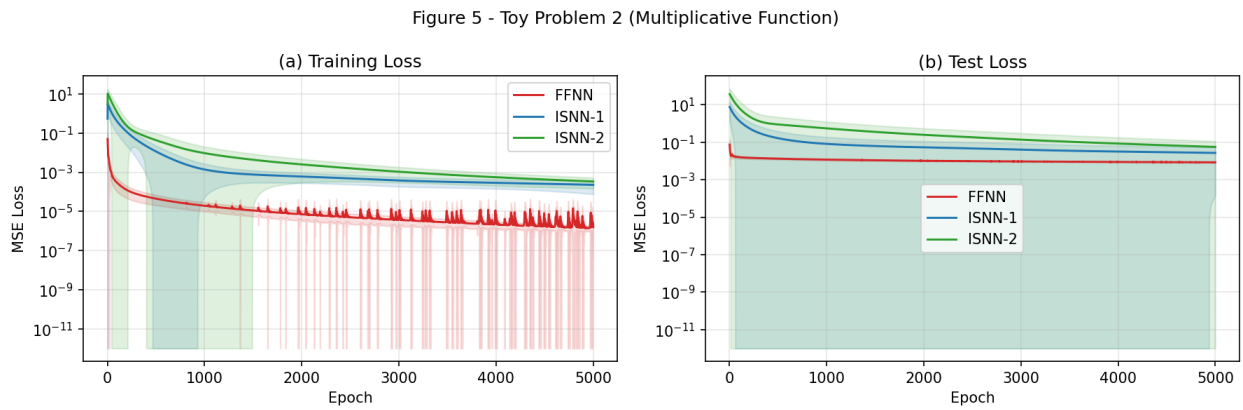
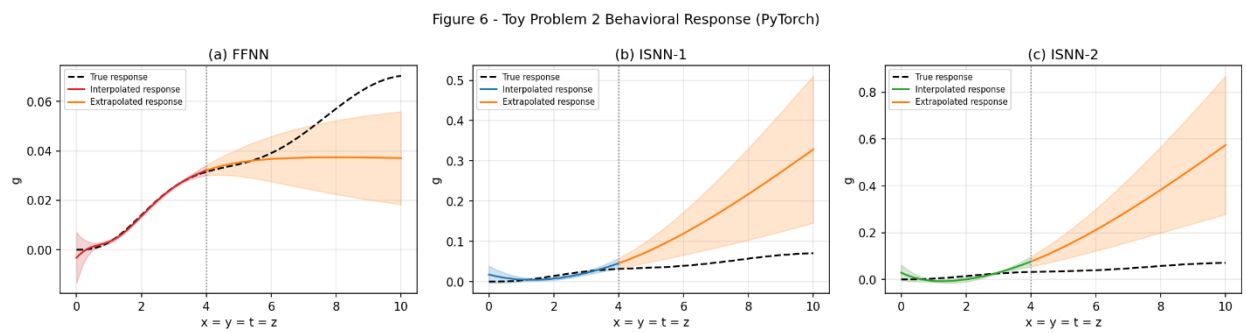


Figure6 Toy1 PyTorch:



PyTorch Toy1 Training:

```
--- Toy 1: FFNN ---
  init 1/10  train=1.095e-04  test=2.185e-01
  init 2/10  train=1.187e-04  test=2.144e-01
  init 3/10  train=1.622e-04  test=2.344e-01
  init 4/10  train=1.243e-04  test=2.148e-01
  init 5/10  train=1.426e-04  test=2.091e-01
  init 6/10  train=9.212e-05  test=1.935e-01
  init 7/10  train=1.000e-04  test=1.967e-01
  init 8/10  train=1.113e-04  test=2.176e-01
  init 9/10  train=1.238e-04  test=2.011e-01
  init 10/10  train=1.116e-04  test=2.507e-01
--- Toy 1: ISNN-1 ---
  init 1/10  train=2.725e-04  test=7.428e-02
  init 2/10  train=1.183e-03  test=7.977e-02
  init 3/10  train=3.414e-04  test=7.328e-02
  init 4/10  train=2.170e-04  test=7.931e-02
  init 5/10  train=3.804e-03  test=5.562e-02
  init 6/10  train=6.238e-04  test=7.638e-02
  init 7/10  train=8.605e-04  test=5.856e-02
  init 8/10  train=1.392e-04  test=8.390e-02
  init 9/10  train=2.609e-04  test=9.588e-02
  init 10/10  train=2.370e-04  test=8.557e-02
```

```
--- Toy 1: ISNN-2 ---
  init 1/10  train=9.311e-04  test=2.288e-01
  init 2/10  train=3.028e-03  test=2.294e-01
  init 3/10  train=1.172e-03  test=2.671e-01
  init 4/10  train=7.874e-04  test=2.228e-01
  init 5/10  train=3.873e-04  test=1.453e-01
  init 6/10  train=1.726e-03  test=2.290e-01
  init 7/10  train=8.108e-04  test=2.318e-01
  init 8/10  train=3.941e-03  test=9.401e-02
  init 9/10  train=1.573e-03  test=1.309e-01
  init 10/10  train=8.436e-04  test=2.180e-01
```

PyTorch Toy2 Training:

```
--- Toy 2: FFNN ---
  init 1/10  train=1.861e-06  test=8.872e-03
  init 2/10  train=1.440e-06  test=8.316e-03
  init 3/10  train=2.286e-06  test=8.947e-03
  init 4/10  train=1.348e-06  test=7.888e-03
  init 5/10  train=1.548e-06  test=8.229e-03
  init 6/10  train=1.319e-06  test=8.694e-03
  init 7/10  train=9.138e-07  test=8.483e-03
  init 8/10  train=1.218e-06  test=8.423e-03
  init 9/10  train=2.938e-06  test=8.802e-03
  init 10/10  train=8.028e-07  test=7.717e-03
--- Toy 2: ISNN-1 ---
  init 1/10  train=5.385e-04  test=2.891e-02
  init 2/10  train=2.537e-05  test=6.949e-03
  init 3/10  train=3.469e-04  test=9.655e-02
  init 4/10  train=2.111e-04  test=1.759e-02
  init 5/10  train=1.033e-04  test=9.747e-03
  init 6/10  train=1.662e-04  test=6.305e-03
  init 7/10  train=5.963e-05  test=6.379e-03
  init 8/10  train=2.078e-04  test=4.485e-02
  init 9/10  train=2.279e-04  test=1.306e-02
  init 10/10  train=3.907e-04  test=3.757e-02
```

```
--- Toy 2: ISNN-2 ---
  init 1/10  train=3.578e-04  test=5.130e-02
  init 2/10  train=2.660e-04  test=1.621e-02
  init 3/10  train=1.592e-04  test=1.863e-02
  init 4/10  train=9.660e-05  test=8.785e-03
  init 5/10  train=3.147e-04  test=3.509e-02
  init 6/10  train=2.118e-04  test=3.212e-02
  init 7/10  train=2.565e-04  test=2.350e-02
  init 8/10  train=3.691e-04  test=8.539e-02
  init 9/10  train=7.497e-04  test=2.119e-01
  init 10/10  train=6.030e-04  test=7.616e-02
```

Manual Training Loop with Adam Optimizer:

Toy1.

```
--- Toy 1: FFNN (Manual) ---
  init 1/10  train=3.571e-04  test=1.737e-01
  init 2/10  train=1.638e-04  test=1.925e-01
  init 3/10  train=3.100e-04  test=2.198e-01
  init 4/10  train=1.859e-04  test=2.107e-01
  init 5/10  train=4.192e-04  test=1.834e-01
  init 6/10  train=2.224e-04  test=1.729e-01
  init 7/10  train=3.922e-04  test=1.407e-01
  init 8/10  train=2.527e-04  test=1.512e-01
  init 9/10  train=3.912e-04  test=1.572e-01
  init 10/10  train=2.816e-04  test=2.067e-01
--- Toy 1: ISNN-1 (Manual) ---
  init 1/10  train=3.123e-03  test=6.492e-02
  init 2/10  train=1.353e-03  test=6.045e-02
  init 3/10  train=1.921e-03  test=7.376e-02
  init 4/10  train=1.937e-03  test=8.502e-02
  init 5/10  train=2.456e-03  test=8.928e-02
  init 6/10  train=5.263e-04  test=1.016e-01
  init 7/10  train=2.885e-03  test=8.190e-02
  init 8/10  train=4.653e-03  test=7.259e-02
  init 9/10  train=7.817e-04  test=6.078e-02
  init 10/10  train=1.894e-03  test=7.054e-02
```

```
--- Toy 1: ISNN-2 (Manual) ---
  init 1/10  train=3.287e-05  test=1.123e-01
  init 2/10  train=7.631e-05  test=4.042e-02
  init 3/10  train=5.035e-05  test=1.531e-01
  init 4/10  train=1.507e-05  test=3.700e-02
  init 5/10  train=3.645e-05  test=1.310e-01
  init 6/10  train=9.074e-05  test=3.448e-02
  init 7/10  train=3.748e-04  test=5.306e-02
  init 8/10  train=4.396e-05  test=1.353e-01
  init 9/10  train=4.116e-05  test=4.100e-02
  init 10/10  train=1.967e-05  test=4.792e-02
```

Toy2.

```
--- Toy 2: FFNN (Manual) ---
  init 1/10  train=2.137e-05  test=1.297e-02
  init 2/10  train=6.741e-05  test=1.648e-02
  init 3/10  train=1.608e-05  test=1.235e-02
  init 4/10  train=3.466e-05  test=1.274e-02
  init 5/10  train=2.740e-05  test=1.127e-02
  init 6/10  train=1.503e-05  test=1.271e-02
  init 7/10  train=1.915e-05  test=1.205e-02
  init 8/10  train=5.595e-05  test=1.295e-02
  init 9/10  train=2.742e-05  test=1.395e-02
  init 10/10  train=1.861e-05  test=1.193e-02
--- Toy 2: ISNN-1 (Manual) ---
  init 1/10  train=4.571e-04  test=2.035e-02
  init 2/10  train=4.942e-04  test=2.113e-02
  init 3/10  train=1.438e-04  test=6.631e-03
  init 4/10  train=2.281e-04  test=9.994e-03
  init 5/10  train=3.748e-04  test=1.740e-02
  init 6/10  train=4.796e-04  test=2.112e-02
  init 7/10  train=2.277e-04  test=9.674e-03
  init 8/10  train=4.292e-04  test=1.944e-02
  init 9/10  train=4.434e-04  test=1.975e-02
  init 10/10  train=3.831e-04  test=1.777e-02
```

```
--- Toy 2: ISNN-2 (Manual) ---
  init 1/10  train=1.526e-04  test=9.831e-03
  init 2/10  train=1.049e-04  test=1.017e-02
  init 3/10  train=1.697e-04  test=1.265e-02
  init 4/10  train=1.645e-04  test=1.161e-02
  init 5/10  train=1.590e-04  test=1.127e-02
  init 6/10  train=1.380e-04  test=1.185e-02
  init 7/10  train=1.491e-04  test=1.077e-02
  init 8/10  train=1.620e-04  test=1.040e-02
  init 9/10  train=1.564e-04  test=1.134e-02
  init 10/10  train=1.518e-04  test=1.003e-02
```

Final Results Summary:

```
PyTorch - Final Mean Test MSE
----------------------------------------
Toy 1  FFNN   : 2.1508e-01
Toy 1  ISNN-1 : 7.6255e-02
Toy 1  ISNN-2 : 1.9971e-01
Toy 2  FFNN   : 8.4371e-03
Toy 2  ISNN-1 : 2.6792e-02
Toy 2  ISNN-2 : 5.5912e-02

Manual NumPy - Final Mean Test MSE
----------------------------------------
Toy 1  FFNN   : 1.8088e-01
Toy 1  ISNN-1 : 7.6087e-02
Toy 1  ISNN-2 : 7.8557e-02
Toy 2  FFNN   : 1.2938e-02
Toy 2  ISNN-1 : 1.6325e-02
Toy 2  ISNN-2 : 1.0993e-02
```

THANK YOU SIR